

# Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 5 ~ BookStream #002.0

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chapter five is called message passing and neurobiology what is your overview thought on chapter five okay uh i mean uh it's a kind of i don't know i have mixed feelings about this chapter because on one hand you see as far as i understand active inference although it originated as a quote-unquote a unified theory of the brain i don't think it's a neurobiological theory per se uh of course there can be some correlations between neurobiological uh components or concepts uh with uh active inference con active inference um i mean concepts but uh i mean it's not uh an essential premise of active inference theory to provide a theory to provide a comprehensive theory about how the neurobiology of human brain or other organisms brain behave uh at a detailed and a neuroanatomical um level uh but at the uh then again it's nice to have these kinds of empirical correlations uh between the findings of neurobiology and the active inference theory uh but i i don't think it's uh one of active inference uh central assertions at least uh to my understanding uh well said very interesting framing well chapter five definitely takes a very specific system of interest approach by highlighting one of the most studied areas also one of the most relevant areas which is mammalian neuroscience and the chapter is going to introduce a few different different motifs in the nervous system and essentially build up towards figure 5.5 which is at the end of the chapter and 5.5 wires together three specific neural systems that the chapter is going to focus on work in that area

from so ali said it very well active inference was built up to in chapter four here is another level or type of science with assertions or with representations or mappings to any specific system but this is the kind of modeling that has been built up and done by friston parr pazulo and others over the decades with a focus coming from a human neural imaging laboratory setting a lot of focus and study and attention and funding and everything on the mammalian nervous system but claims about the nervous system have have uh uh

are not the basis of what active inference claims or how it's derived but this is like an example case study in neurobiology connecting back to some of the formalisms that we've just seen introduced in chapter four yeah and uh to add a minor point uh to which you just said i think it's important to uh draw attention to uh the last sentence of uh the last paragraph of the first page it is important to draw a distinction between a principle i.e the minimization of free energy and a process theory about how this principle may be implemented in a certain kind of system so i i think uh this sentence here uh frames this chapter uh in relation to

all the other technical chapters of this book so uh if uh every other chapter is about uh developing or at least up to now was about developing um the principled formalism of active inference now chapter five provides uh a kind of preliminary sketch for uh the process theory of active inference which is uh obviously far from an extensive theory it's just a single chapter but then again uh it can um provide some important uh

signpost signposts for anyone who wants to uh further investigate uh this area awesome free energy principle bayesian mechanics all things in that area are on this principle not responsive to empirical data data and then the process theory is about how the principle is implemented so the specific generative models that are made and how well they map or how well they do in a portfolio of models that can have very different goals and assumptions and all of this but the process theory implementation lets us develop hypotheses that are answerable to empirical data like what is the kind of information information or relationship between photons hitting the retina and changes in activity in neural systems and that's an informational question or can be abstracted in a way to an informational question that it turns out does have empirical support and results in unique explanations and predictions that doesn't mean that it always results in unique explanations and predictions but a lot of citations are provided here that's what we can explore in chapter five the last paragraph of the first section describes that they're going to look at the three different neural systems

uh okay section 5.2 micro circuits and messages

what do you think ali uh all right so uh i mean this chapter begins uh from how message passing happens uh in neurobiological terms

uh and compare it to the way active inference frames this message passing mechanism and specifically if we look at uh figure 5.1

uh and uh compare this figure to uh the ones we've seen uh before uh in chapters uh uh in chapters uh uh uh in one through four i think it was in chapter four uh we can see some clear parallels between how this kind of uh cortical message

uh how this kind of passage uh happens uh happens in the brain versus how it is framed in active inference literature and uh as we can see it's uh

uh clearly inspired by the neurobiology of the brain uh but uh then it's important to keep in mind that it's not a direct one-to-one mapping between uh these two models uh this is just um

uh this is just um a kind of i don't know uh an interesting or illuminating uh if you like parallel to keep in mind uh to somehow be uh a bit more confident about uh the viability of the theory we want to use for message passing in active inference uh

uh which is to say it's uh which is to say it's not some haphazard theory that's just been developed for practical reasons uh it has some basis in neurobiology although uh it's not necessarily fully uh congruent with um every detail of neurobiology neurobiology

great the specific example is going to involve this one region of mammalian cortex tissue that has these six layers and there's a ton of neurobiology

the big takeaway for figure 5.1 is that it's possible to graphically lay out nodes and variables and find some empirical correspondences again some unique explanations and predictions in certain cases and that's one kind of modeling where it's really trying to understand uh and improve the ability to do correlation and intervention and counterfactual causal type analysis with the real system of interest or in a more pedagogical setting or research setting or an industrial setting you might sweep across large families of structures of models and there's no need to be grounded to any biological structure at all so this is just describing the specific neuro anatomical research that really arose out of the imaging work

at ucl and spm package so that's where a lot of this comes from 5.2 yeah good and sorry just as a side note uh i think

watching uh one of thomas parr's lectures on neurobiology of active inference which is available on youtube would really help to understand the materials of this uh chapter better so i highly recommend watching that one

thanks figure 5.2

gives a re-rendering of a kind of classical view of a hierarchical predictive coding system works so here abstracting a layer from the tissue six layer to just two layers here

the computational layers now and then showing how there's hierarchical communication within a layer but also uh there's there's signaling within a layer and there's a hierarchy uh in bayesian modeling with variables that are higher order predictions about other variables and that's the basis of the predictive coding architecture so 5.2 looks at some ways that the something that resonates with the cerebral cortical architecture enables what might computationally look like or have some really strong and explanatory values in actually relating to computationally a hierarchical bayesian model which could do various general tasks all right 5.3 is motor commands leaving the prefrontal cortex going down to the butterfly looking cross section here what is 5.3

okay so uh 5.3 moves to the other half of active inference framework which is uh i mean how how it can model

[illegible]

particularly related to I mean the motor commands and how it can relate to active inference and particularly the continuous time active inference so we can see that for I mean for the external event or I'm sorry for the external state we can take for example the proprioceptive afferent and then this proprioceptive afferent acts as a kind of why for the continuous time active inference which needs to be I mean processed in in a way to optimize the expected free energy and how it relates to both attention and precision we will see a bit more detail about those those terms and the relation between them in chapter eight but I think here section 5.3 provides a good summary about the general

uh paths through uh the motor uh command systems of neurobiology great I'd say while the previous case study focused on how the connectivity within in between the cortical columns could have a computational relationship with a Bayesian hierarchical predictive coding architecture the argument of the second case study is that a continuous input continuous output kind of set point seeking

reflexive motor behavior with a moving set point with a descending moving set point enabling motion by changing ultimately the set point and enabling uh variation in the strategies to reach that set point through different mechanisms this is also describable in a compatible way

that's a shorter section now section 5.4 subcortical structures

what would you say about this section

okay so uh subcortical um structures are uh very important in uh the the decision making and um I mean uh the planning uh of the agents so uh obviously here uh we need to uh uh we need uh another kind of analogy uh between uh the way uh that these plannings and uh decision makings uh happen uh neuroanatomically uh with

the way uh that uh that it's framed in active inference but again uh we can see it's uh clearly based on uh

I mean at least some of the important elements we've seen uh from uh the previous chapters uh so for example we we saw how

uh policy is described or how uh how it relates to outcomes and uh and preference and so on uh we can see uh

those elements are directly inspired by a neuroanatomical structures uh so uh

I guess that's uh uh at least in my opinion uh this section here 5.4 uh seems a bit more uh

sketchy in the meaning that it doesn't go into quite uh the extensive details uh about how how how those

structures can be um can be compared but uh for anyone who wants to uh further uh investigate uh these

topics there are some uh useful references uh put on here uh pages uh 93 and 94. so yeah thanks yeah it's

really abbreviated and overviewed but we get an interlude from table 5.1 with putative roles of

neurotransmitters so same perspective that we took before on neuroanatomical functionalism

here directly translates to neurotransmitter reductionism or essentialism or something like

that so certainly all neurotransmitters and molecules they play variable roles in different settings and

this is the neat and scruffy

manifold all over again

one person might say well we we need a theory for every acetylcholine molecule in the world they're

all in a unique context and someone else says all neurotransmitters are described by one

parameter in this model i'm getting value from it so to me that's an account and uh somewhere in between

is the work in this space which is making uh in an attempt to have a principled and falsifiable

approach to model the computational aspects of specific regions and contexts and settings

and so acetylcholine noradrenaline dopamine and serotonin are given a little mini review here

and so it's not an exhaustive or an exclusive claim it's kind of a provocation from computational and

molecular neuroscience and people can look into the papers and also ones that probably have been published

since

5.6 goes to continuous and discrete hierarchies which is graphically overviewed in figure 5.5 so what

would you say about this

yeah one interesting thing about this section is um the observation that our lower level

uh engagement with the environment uh it can be most successfully uh characterized with continuous time formulations but as we go up uh on uh i mean on the level of uh uh cognitive concepts or at the level of cognitive hierarchies and uh we come to uh concepts such as i don't know decisions or uh even beliefs and so on

uh we can reach um the area that uh the discrete time situations uh would probably be more efficient to uh to characterize uh the behavior of the agent uh so this multi-scale um

multi-scale structure of active inference modeling is quite evident uh in the way that uh our message passing uh happens in our brain in terms of our lower level data processing uh up into consolidating uh the higher level cognitive concepts and ontologies awesome thank you

to me figure 5.5 demonstrates the kind of whole of body approach

that you could imagine there's so many organs and systems and phenomena for which there aren't specific generative models so little can be said about situations where no generative model

has been articulated and here's one where it has so it gives you also it's kind of like reading a

drosophila melanogaster review paper relatively it's like this is how much work it takes to get to this

state of knowledge in an insect so then in another insect do we know less about that insect empirically

and genetically so consider this to be what's known to be a lot however also about one of the most

sophisticated or specific cognitive systems at least we know so there's that additional kind of like

self-reflexive um aspect to this chapter that is not a cornerstone of active inference but here it's just

presented in a synthetic case study anything else you want to say about five uh nothing particular comes to mind

all right